Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Research & Open-Source Experience

Core Contributor, RookDB Aug 2026 – Present
Independent Study, DSAC Lab, IIIT Hyderabad | Rust, SQL, Databases *Hyderabad, India*

- Co-designed the lab's open-source SQL database in Rust: storage engine, buffer pool, B+Tree indexes and query executor.
- Built the on-disk page storage and a shared buffer pool that keeps frequently used pages in memory.
- Cut memory allocations in the query engine by processing rows in batches and reusing buffers.
- Led performance work that made the benchmark suite **14x faster**, backed by **1,200+ tests**.

Projects

Warehouse Traffic Control: Fault-Tolerant Robot Coordination | *JavaScript, Web Workers*

- Designed a lease-based protocol where 16 managers coordinate 200 robots with **zero collisions** over 416K moves.
- Added deadlock detection and crash recovery, so a failed manager rebuilds its state in seconds.
- Compared with a central coordinator: **7.6x lower** peak load per node, and a crash affects 11 robots instead of 120.

Peer-to-Peer Distributed File Sharing System | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

FOOL: Custom POSIX-Style Interactive Shell | *C++, Linux, System Calls, Signals*

- Built a Unix shell in C++ directly on Linux system calls, with pipelines, redirection and job control.
- Fixed race conditions in signal handling so the shell never loses track of finished processes.
- Used `strace` to cut system calls in the line editor, and made startup **52x faster**.

Technical Skills

Languages: C/C++, Rust, Python, Bash, SQL, JavaScript.
Linux: Processes, Signals, Pipes, Threads, Sockets.
Tools: Make, Git, strace, ASan/UBSan, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.